

Paper 5

AI-Assisted Portfolio Triage for Canada's Major Projects: Power Ranking and Competitive Funding Allocation Use Case

Running Title: AI Portfolio Triage – Power Ranking

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Abstract

Canada's Major Projects Inventory (MPI) lists more than 500 large-scale energy, mining, and forestry proposals but offers little guidance on which deserve priority. Earlier work developed a Bayesian likelihood-ratio scorecard to estimate build probability and a Cox proportional hazards model to estimate time-to-event. This paper introduces power ranking, a composite score that integrates feasibility and urgency into a single, auditable metric. Power ranking enables objective comparison of projects even outside direct competition, generating transparent queues for allocating resources within competitive envelopes. The Clean Fuels Fund ("CFF") illustrates the approach. Filtering the 2024 MPI for CFF eligibility, power ranking highlights 11 capital projects and 12 feasibility proposals, together accounting for 96 percent of the fund's \$500-million envelope. Projects with mid-range probabilities but urgent timelines rise to the top, showing how power ranking sharpens triage beyond probability alone. The framework equips developers, planners, financiers, and policymakers with a reproducible tool for sequencing portfolios, designing competitive rounds, and directing scarce resources toward projects most likely and most urgent to advance.

Introduction

Canada's Major Projects Inventory (MPI) provides the most comprehensive public record of large-scale energy, mining, and forestry developments. Yet decision makers continue to face a central challenge: how to compare projects objectively when they are not competing head-to-head in a single process but instead emerge in fragmented regulatory, financial, or policy pipelines. In these contexts, projects must be assessed not only on their individual merits but also on their relative strength across a portfolio.

Papers 1 and 2 examined the MPI historically, identifying advancement patterns, bottlenecks, and attrition rates. Paper 3 introduced a pair of lightweight, auditable models: a Bayesian likelihood-ratio scorecard estimating build probability, and a Cox proportional hazards model estimating time-to-event, which achieved functional accuracy and interpretability. Paper 4 operationalised these risk engines at the portfolio level, applying them to the 2024 MPI cohort to produce blended probabilities and an urgency-adjusted priority index, together forming a triage matrix for resource allocation.

This fifth paper advances the framework by introducing power ranking as a way to consolidate these metrics into a single scoring formula. Power ranking combines blended probability with the priority index to produce an integrated measure of both feasibility and timing. The innovation addresses a persistent knowledge gap: existing approaches either rely on top-down techno-economic heuristics, which flatten project-level differences, or on expert judgment, which drifts over time and lacks transparency. By contrast, power ranking provides a reproducible, algebraic method for generating transparent, data-driven queues that can be applied across programs, funding envelopes, or investment portfolios. It enables projects to be compared objectively even

when they have not directly competed against each other, offering a nuanced view of relative strength within a diverse pipeline.

The Clean Fuels Fund (“CFF”) serves here as a use case. With its \$500 million envelope and open-intake design, CFF applicants face the dual challenge of demonstrating feasibility and urgency in a crowded field. Applying power ranking to the MPI filters, eligibility screens, and model outputs produces a defensible shortlist of projects likely to deliver clean fuels by 2030, while also illustrating how the same approach can guide other grant programs, competitive envelopes, or private capital allocation.

In this way, Paper 5 builds directly on the statistical foundations established in earlier work and extends them into a practical tool for decision making. Power ranking translates the MPI from a dataset into a comparative benchmark, allowing developers, planners, funders, and policymakers to sequence proposals with greater confidence.

Methods

Data foundation

The analysis draws on the 2024 edition of the MPI, which lists 504 large energy, mining, and forestry proposals worth roughly \$650 billion. To qualify for inclusion in the MPI, a project must exceed the relevant capital-cost threshold: \$50 million for energy and mining, \$20 million for electricity and forestry, and \$10 million for clean-energy and clean-technology initiatives. For every project, the MPI records a status code such as Announced & Planning, In Review, Approved, or Under Construction, allowing year-by-year tracking of progress.

Screening for Clean Fuels Fund eligibility

The CFF requires projects to produce liquid or gaseous clean fuels and to be capable of commissioning before 2030. The MPI entries were filtered accordingly. Hydrogen proposals were automatically routed to the feasibility stream, since the program supports design work but not construction for pure hydrogen projects. Solid biomass projects were excluded, as the CFF general stream funds only liquid or gaseous fuels. Terms such as renewable natural gas, ethanol, renewable diesel, and methanol were mapped directly to CFF categories, while ambiguous descriptors were manually reviewed and classified.

Eligibility screening then applied the CFF's cost-share and scheduling rules. Capital projects could receive the lesser of \$50 million or 30 percent of total cost, if construction could begin by 2028 and reach operation by 2030. Feasibility studies were budgeted at 3 percent of project cost, with the CFF share capped at \$5 million per study.

Probabilistic modelling

Prior work (Papers 3 and 4) established two complementary risk engines. The first is a Bayesian likelihood-ratio scorecard, which begins with a historical prior of 28 percent of projects reaching construction within five years and updates those odds with likelihood ratios derived from project descriptors such as province, sector, cost, and cleantech status. Unknown values revert to a neutral prior. The second is a Cox proportional hazards model, calibrated on the 2018–2020 MPI cohort, which estimates time-to-event probabilities using covariates expressed as hazard ratios. For the three-year horizon, a fixed baseline survival function of $S_0(3) = 0.9491$ was applied.

Each model produces a project-level probability: the Bayesian scorecard yields p_{bayes} , the Cox model yields p_{cox} . These are blended at a 60–40 weighting to form a composite three-year build probability, $blended_prob$. In parallel, a priority index is computed as $blended_prob$

divided by years remaining to the expected five-year median decision horizon, floored at 0.25 years.

Power ranking formula

The innovation of this study is the introduction of power ranking as a shorthand for a composite scoring formula that integrates probability and urgency into a single measure. Power ranking is defined as:

$$\text{Power Ranking} = 0.60 \times \text{blended_prob} + 0.40 \times \text{priority_index (rescaled 0–1)}$$

This formulation ensures that projects with both high probability and pressing timelines rise to the top of the queue, while projects with weaker fundamentals or long horizons fall lower. The weighting was chosen to emphasize overall feasibility while retaining urgency as a decisive factor. Sensitivity tests confirmed that alternative weights in the 50/50 to 70/30 range did not materially change the funded set of projects.

Application to the Clean Fuels Fund

For capital projects, CFF allocations were simulated by ranking eligible candidates in descending order of power ranking and awarding support until 90 percent of the \$500 million envelope was committed, leaving flexibility for feasibility awards. Feasibility projects were evaluated separately under program rules, with CFF contributions capped at \$5 million each. Hydrogen projects were automatically routed to this stream, as were two renewable natural gas projects whose sponsors elected to complete front-end engineering before seeking construction finance.

Quality assurance

Several safeguards were built into the modelling process. Input files were checked for column completeness and correct dummy encodings. Posterior probabilities were validated to remain strictly between zero and one. Composite rankings were cross-checked to ensure that rescaling of the priority index preserved order. All coefficient files were hash-logged to allow external replication.

Results & Discussion

Eligible project pool

Filtering the 2024 MPI for CFF eligibility produced 34 candidate projects across six provinces, representing \$22.7 billion of potential investment. Project sizes ranged from \$11 million to \$5 billion, with a mean of \$669 million and a median of \$127 million. British Columbia contributed 15 proposals, Alberta nine, and the remainder were spread across Quebec, Saskatchewan, Manitoba, and Ontario. Hydrogen accounted for ten projects, biomass-to-liquid or gas for six, and renewable natural gas for five.

Figure 1 presents a visualization dashboard of the 34 candidates through six interconnected charts. Provincial distribution highlights British Columbia's 15 projects and Alberta's nine. The fuel-type breakdown shows hydrogen's dominance, while cost histograms reveal a skewed pipeline ranging from small ventures under \$100 million to megaprojects above \$1 billion. The blended probability distribution indicates that 17 projects clear a 50 percent threshold, while 15 fall below 10 percent. Finally, a scatter plot of project cost versus composite scores demonstrates that strong candidates appear across all cost tiers, though high-scoring projects cluster tightly at the top.

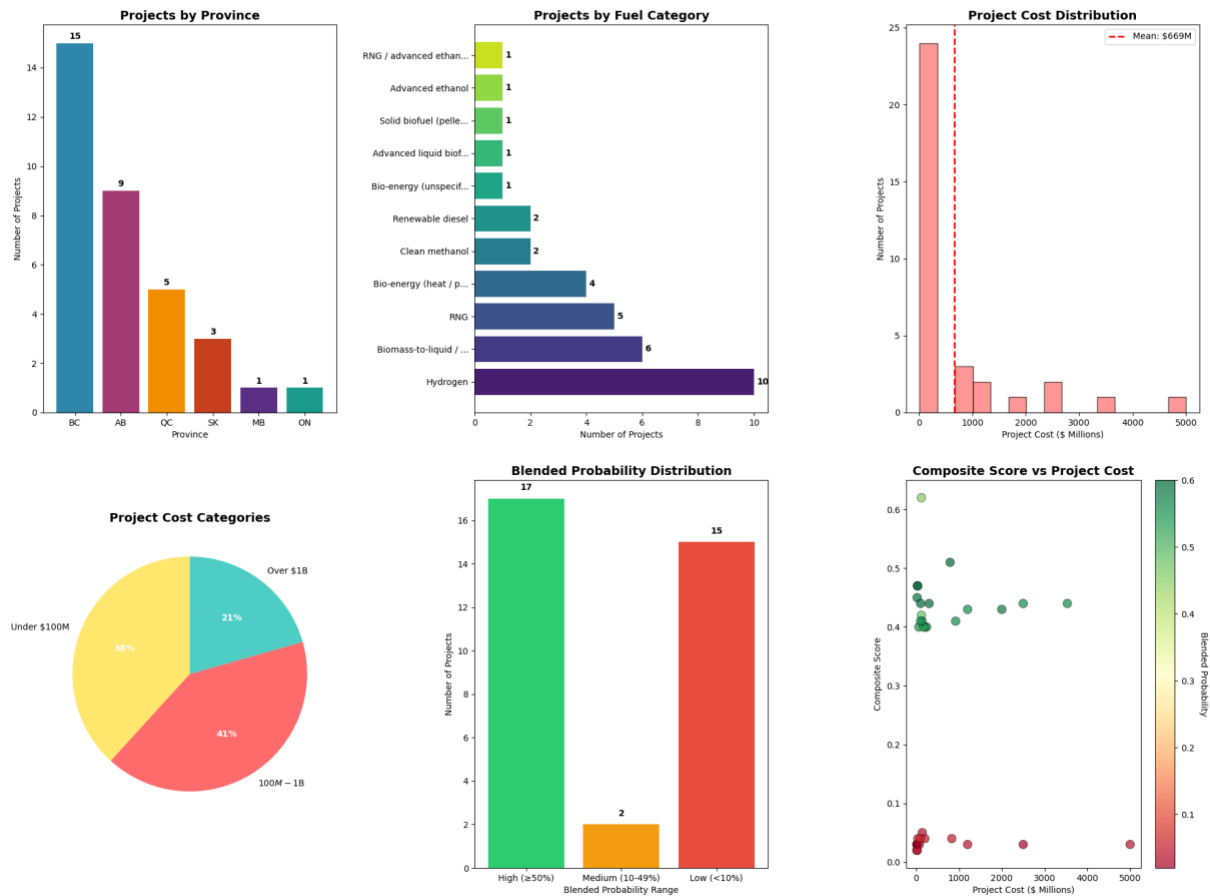


Figure 1: Visualization of CFF Eligible Projects.

Distribution of probabilities and priority

The Bayesian scorecard generated probabilities spanning the full range, with many projects clustering near zero or above 50 percent. The Cox model was more conservative, rarely exceeding 10 percent. Blending the two produced mid-range values centered near 32 percent.

The priority index introduced sharper contrasts. While most projects fell into a low-urgency band, a handful scored above 1.5, signalling imminent decision points. Combining probability and priority into a power ranking produced a polarized distribution: 19 projects achieved scores above 0.40, while 15 fell below 0.10.

Allocation results

The modeled allocation of Clean Fuels Fund support is presented in Tables 1 through 3.

Together, the tables show how power ranking nearly exhausted the envelope while remaining consistent with program rules.

Table 1 shows the capital stream. Eleven projects rose above the funding line, securing about 90 percent of the overall CFF envelope. These awards concentrated on proposals that combined mid-range probabilities with elevated urgency. Alberta captured six of the 11 funded projects, while Saskatchewan and Quebec each secured two and Ontario one.

Unique ID	blended prob	priority index	power ranking	project cost (C\$MM)	CFF capital (C\$MM)
AB					
6002	0.60	0.30	0.43	\$25	\$8
6067	0.56	0.28	0.40	\$2,500	\$50
6088	0.60	0.30	0.43	\$35	\$11
7035	0.56	0.19	0.38	\$1,200	\$50
8017	0.60	0.15	0.39	\$300	\$50
8038	0.60	0.15	0.39	\$100	\$30
SK					
7029	0.58	0.19	0.39	\$2,000	\$50
8319	0.56	0.14	0.37	\$920	\$50
QC					
7009	0.54	0.18	0.36	\$243	\$50
7086	0.54	0.18	0.36	\$200	\$50
ON					
9001	0.53	0.18	0.35	\$170	\$50
Grand Total	0.57	0.20	0.38	\$7,693	\$448

Table 1: Power Ranking for CFF Capital Stream.

Table 2 lists the feasibility stream. All 12 eligible MPI projects received support totalling \$34 million, approximately 7 percent of the envelope. Hydrogen dominated this stream with ten awards, joined by two renewable natural gas projects whose sponsors opted to complete design work before construction finance.

Unique ID	blended prob	priority index	power ranking	project cost (C\$MM)	CFF feasibility (C\$MM)
MB					
7038	0.59	0.20	0.40	\$20	\$0.5
AB					
6091	0.56	0.28	0.40	\$3,532	\$5.0
9002	0.56	0.14	0.36	\$135	\$3.0
SK					
9003	0.55	0.14	0.36	\$100	\$2.3
BC					
7031	0.04	0.01	0.03	\$1,200	\$5.0
7039	0.03	0.01	0.02	\$11	\$0.3
7044	0.05	0.02	0.03	\$200	\$4.5
7059	0.04	0.01	0.03	\$5,000	\$5.0
8065	0.05	0.01	0.03	\$830	\$5.0
8068	0.06	0.01	0.04	\$36	\$0.8
8069	0.06	0.01	0.04	\$90	\$2.0
8071	0.03	0.01	0.02	\$22	\$0.5
Grand Total	0.22	0.07	0.15	\$11,176	\$33.8

Table 2: Power Ranking for CFF Feasibility Stream.

Table 3 summarizes the overall fiscal position. Total commitments of \$482 million left \$18 million unallocated, sufficient for one or two small capital projects or several additional feasibility studies.

Row Labels	count	blended prob	priority index	power ranking	project cost (C\$MM)	CFF (C\$MM)
CFF Capital Stream	11	0.57	0.20	0.38	\$7,693	\$448
CFF Feasibility Stream	12	0.22	0.07	0.15	\$11,176	\$34
Grand Total	23				18,869	482

Table 3: Consolidated CFF Contribution Envelope.

Insights from power ranking

The CFF case illustrates how power ranking consolidates probability and timing into a transparent scoring system that can order projects even outside a direct competition. Three insights stand out. First, funded projects clustered in the middle of the probability distribution, around 50 percent blended probability, basically a coin toss. Second, the \$50 million contribution

cap broadened the portfolio, ensuring that smaller, higher-ranked projects were not crowded out by mega-projects. Third, the priority component of the power ranking proved material: without it, projects under acute time pressure could have fallen below the funding line.

Implications for developers

For developers, power ranking offers a benchmark for understanding how their projects stack up against peers. In the CFF case, the ranked list clarified whether a proposal should proceed as a capital application, be redirected to feasibility, or be deferred. Projects that landed in the upper tier combined probability and urgency in a way that justified advancing a full application despite the administrative burden. Conversely, proposals with very low power ranking could be deprioritized, conserving scarce internal bandwidth. Beyond CFF, the same approach can guide project sponsors in other grant competitions, provincial programs, or even internal capital budgeting.

Implications for policymakers and program staff

For program administrators, power ranking provides a transparent, auditable method for allocating scarce funds. In CFF, the scoring formula reproduced the program's dual mandate: capital grants flowed to projects with both strong odds of success and time-sensitive milestones, while feasibility support was directed toward hydrogen and other early-stage files. More broadly, the same framework could underpin future envelopes, such as federal clean power credits, provincial innovation funds, or regional infrastructure programs, by generating defensible queues that refresh automatically as inputs update. This improves consistency and reduces the risk of discretionary bias.

Implications for financiers

For financiers, the power ranking tool reframes how to triage competing opportunities.

Traditional project finance assessment already weighs feasibility and timing, but rarely in a single transparent metric. The CFF results show that projects around the 50 percent probability mark, when paired with pressing deadlines, can be more compelling than projects with higher standalone feasibility but slower clocks. For lenders and investors, this distinction can shape structuring terms, hurdle rates, and the sequencing of due diligence across a portfolio.

Broader applicability

The broader value of power ranking lies in its ability to compare projects that do not directly compete against each other. In most policy and investment contexts, proposals arrive asynchronously, are sponsored by different entities, and fall under multiple jurisdictions. By collapsing probability and urgency into a single algebraic score, power ranking enables apples-to-apples comparison across this diversity. This transforms a static registry like the MPI into a living decision-support tool, equally useful for public funders allocating finite grant envelopes, regulators sequencing reviews, or companies prioritising internal capital deployment.

Limitations

Several caveats temper these findings. First, the MPI snapshot fixes project descriptors at mid-2024, meaning that subsequent cancellations or new announcements are not captured. Second, the models assume independence among Bayesian predictors and proportional hazards in the Cox specification, both simplifications. Third, the CFF-specific mapping exercise required judgement calls where MPI descriptors did not map cleanly to program categories. Despite these

limitations, the analysis demonstrates that power ranking produces stable, reproducible signals that align with observed program rules and realistic funding outcomes.

Conclusion

This study introduced power ranking as a practical method for consolidating probability and urgency into a single, auditable score for project triage. Building on the Bayesian scorecard and the Cox proportional hazards model developed in earlier work, power ranking integrates the blended probability of construction with a rescaled priority index to generate transparent, data-driven queues.

The CFF served as a use case. Applying power ranking to the 2024 MPI filtered 504 projects to 34 potential candidates. Of these, 14 met all capital-stream criteria and 11 capital projects were selected before the envelope reached its threshold, absorbing 90 percent of available funding. Beyond the CFF, the approach has broader utility. Developers can benchmark their projects against a shifting pipeline and allocate internal resources accordingly. Policymakers and program staff can deploy an auditable ranking system to distribute competitive envelopes consistently, reducing the risk of bias and improving portfolio balance. Financiers can use the same score to differentiate between projects with similar technical merit but divergent timing pressures, aligning structuring terms more closely with risk.

Limitations remain. The MPI snapshot fixes project descriptors in time, and the models rest on simplifying assumptions about predictor independence and hazard proportionality. Eligibility mapping also required judgment where program categories diverged from MPI labels. Yet despite these caveats, power ranking demonstrates stability across sensitivity tests and consistency with real program constraints.

By extending probability and urgency into a unified metric, power ranking closes the gap between risk assessment and resource allocation. It enables projects to be compared even outside direct competition, creating defensible queues for developers, funders, and policymakers. In doing so, it transforms Canada's static project registry into a dynamic triage tool that can guide decision making across clean energy programs, infrastructure funds, and private capital portfolios.